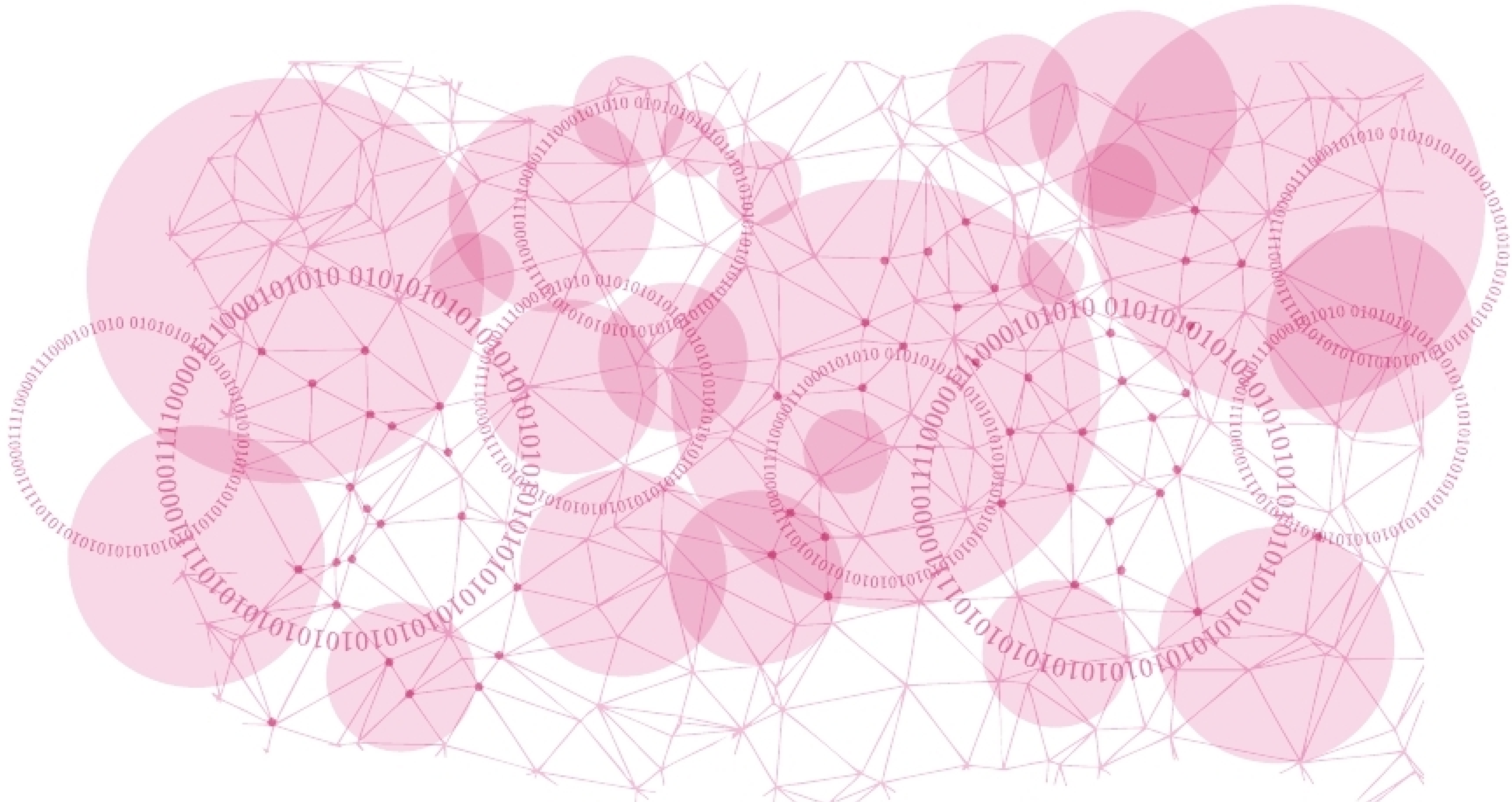
Processing: A Programming Language for Animation

Here's an overview of the Processing programming language, and a tutorial to demystify the process of building a simple animation with it.



In the digital era, animation can do more than just entertain children. It's an effective tool for visual communication. Of course, it offers a whole new medium for expression and creativity, but at a more practical level, the movement in an animation attracts more attention than static images.

Programming can be fun and a very creative activity. Playing with graphics and sound is a great way to start programming. Let's answer three simple questions to get started: What is Processing, who should use it, and why should they do so?

The Processing programming language

Processing is an open source graphical library and integrated development environment built for the creation of digital arts and multimedia, with the purpose of teaching non-programmers the fundamentals of computer programming in a visual context.

It is based on Java and all the code gets pre-processed and converted directly into Java code when you hit the *Run* button. Java's *PApplet* class is the base class for all Processing sketches.

Processing is easy to learn, without having to type a lot of code. It is so simple that you can read through the code and get a rough idea of the program flow.

Who can use Processing?

Processing is designed not only for coders who want to create interactive visual programs but can also be used by designers and artists with minimal, if any, programming experience. If that sounds interesting to you, then welcome to Processing!

The features of Processing are:

- Free to download and is open source
- Interactive programs with 2D, 3D, PDF, or SVG
- Easy to learn and understand
- Available for GNU/Linux, Mac OS X, Windows, Android and ARM
- Over 100 libraries to extend the core software

Processing consists of:

- *Processing Development Environment (PDE)*: The PDE is an Integrated Development Environment (IDE) with a minimalist set of features.
- A collection of functions (also referred to as commands or methods) that make up the 'core' programming interface.
- A language syntax, identical to Java but with a few modifications.